llmXive Automated Review of OmniRetrieval: Unified Retrieval across Heterogeneous Knowledge Sources

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The primary concern regarding claim accuracy is the citation and existence of the backbone models used for evaluation. The manuscript repeatedly cites “GPT-5.4” (e.g., Table 1, Section 5.2) and “Gemini-3.1” as the specific models used to generate the reported results. However, these model versions do not exist in the public domain, nor are they supported by the provided bibliography. The bibliography lists `GPT-5` (arXiv:2601.03267) and `Gemini-2.5` (arXiv:2507.06261), but the text refers to “GPT-5.4-mini” and “Gemini-3.1 (Pro)”. Furthermore, the arXiv IDs provided in the bibliography point to future dates (2026), suggesting the paper may be a synthetic or hallucinated artifact rather than a real preprint. If the models cited (GPT-5.4, Gemini-3.1) are fictional, the entire experimental section, including the quantitative claims of outperforming baselines, is unsupported by evidence.

Additionally, the citation keys in the text (e.g., `\citep{GPT-54}`) do not align with the semantic names used in the prose (“GPT-5.4”), and the bibliography entry `GPT-54` is ambiguous. The claim that the system was evaluated on “309 distinct knowledge bases” is mathematically consistent with the sum of the components (206+80+1+15+7), but the treatment of the 7 BEIR datasets as single “knowledge bases” rather than distinct corpora within a single benchmark suite is a minor semantic inaccuracy that slightly inflates the perceived diversity of the structured data sources. However, the non-existence of the primary evaluation models is a fatal flaw that invalidates the paper’s core empirical claims.
Action items.

- [fatal] The paper cites ‘GPT-5.4’ and ‘Gemini-3.1’ as evaluation backbones. These models do not exist in public records or the provided bibliography (which lists GPT-5 and Gemini-2.5). This appears to be a hallucinated citation, rendering the experimental results unverifiable.
- [fatal] Citation keys like ‘GPT-54’ in the text do not match the semantic names ‘GPT-5.4’ used in prose. The bibliography entries point to future dates (2026) or mismatched versions, undermining the claim of using specific, real-world backbones for the reported metrics.
- [writing] The claim of ‘309 distinct knowledge bases’ treats 7 BEIR datasets as single bases. While mathematically summing to 309, this phrasing is misleading as it implies 309 separate database instances rather than distinct corpora within a benchmark suite.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 is a clear and well-structured conceptual diagram that effectively illustrates the problem of heterogeneous retrieval sources (left) and the proposed OmniRetrieval solution (right). The visual elements, including icons, colors, and flow arrows, align perfectly with the provided caption, and the internal labels are legible and sufficient to understand the architecture without needing external definitions.

Figure 2

The figure is a stacked bar chart comparing source selection outcomes, but it lacks a caption entirely, obscuring its scientific context. Additionally, the data bars do not sum to 100%, and the legend is partially covered by the title.

Figure 3

The figure effectively visualizes the distribution of predictions across different query types and models, but it suffers from a complete lack of a descriptive caption, leaving the context and specific definitions of the reference line to the reader's inference.

Figure 4

The figure presents clear scaling trends across three tasks, but it is missing a descriptive caption and the legend is not explicitly repeated or referenced for the first two subplots.

Figure 5

The figure is visually clear with distinct series and annotations, but the complete absence of a descriptive caption makes it impossible to verify the experimental context or the specific meaning of the reported accuracy metrics.
Action items.

- [fatal] Figure 2: The figure has no caption (stated as '(no caption)'), making it impossible to verify if the displayed data matches the authors' intended claims or context.
- [science] Figure 2: The stacked bars do not sum to 100% (e.g., the 'GPT-5.4' bar sums to ~98.9% and 'Gemma-4' sums to ~99.5%), suggesting missing data categories or calculation errors.
- [writing] Figure 2: The legend is partially obscured by the figure title, making the '2-Cand., Gold Included' entry difficult to read.
- [writing] Figure 3: The figure lacks a descriptive caption; the current placeholder '(no caption)' fails to explain the experimental setup, the meaning of the 'Balanced Reference' line, or the specific models being compared.
- [writing] Figure 3: The legend is positioned outside the plot area and is not enclosed in a box, which can make it visually disconnected from the data in some rendering contexts.

- [writing] Figure 4: The figure lacks a descriptive caption, providing no context for the 'Source Selection', 'Retrieval', and 'Judge' subplots or the specific experiment being visualized.
- [writing] Figure 4: The legend is located only in the rightmost subplot ('Judge'), making it ambiguous for the 'Source Selection' and 'Retrieval' subplots which use the same symbols but lack a direct legend.
- [writing] Figure 5: The caption is missing; the provided text '(no caption)' prevents verification of the specific experimental setup, dataset, or metric definitions required to interpret the 'Accuracy' and 'Candidate Sources' axes.
- [writing] Figure 5: The y-axis label 'Accuracy (%)' is ambiguous without a caption specifying the exact task (e.g., retrieval, generation, or classification) or the baseline against which the 'Oracle' is compared.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on specialized terminology that obscures meaning for non-expert readers. The most frequent offender is "structural affordances," which appears in the Abstract, Introduction, and Related Work without ever being defined. While the authors imply it refers to schemas or query capabilities, a general reader cannot parse this without guessing. Similarly, "native execution engines" (Section 2.1) and "atomic-unit retrieval" (Section 5) are jargon-heavy phrases that could be replaced with "databases," "query processors," or "single-source retrieval" respectively.

The term "verbalized" in Section 2.4 is used to describe converting structured results (rows, triples) into text, but this is not standard terminology outside of specific NLP subfields; "converted to text" or "summarized" would be clearer. Additionally, "backbone" is used repeatedly to refer to the underlying LLM (e.g., Section 3.1), which is acceptable in the field but should be explicitly defined as "base model" or "underlying LLM" upon first mention to aid broader comprehension.

Finally, "modality gap" (Section 1) is used to describe a bias in retrieval but lacks a concrete definition. The paper assumes the reader understands that this refers to the mismatch between query form and source structure. Replacing these terms with plainer language or adding brief parenthetical definitions would significantly improve accessibility without sacrificing precision.

Action items.

- [writing] The manuscript relies heavily on specialized terminology that obscures meaning for non-expert readers. The most frequent offender is "structural affordances," which appears in the Abstract, Introduction, and Related Work without ever being defined. While the authors imply it refers to schemas or query capabilities, a general reader cannot parse this without guessing. Similarly, "native execution engines" (Section 2.1) and "atomic-unit retrieval" (Section 5) are jargon-heavy phrases that could be

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The logical consistency of the paper is generally sound in its high-level narrative: heterogeneous sources require native query languages, and a unified representation flattens structural affordances. The experimental design largely follows this premise. However, there are specific inconsistencies in the interpretation of results and the presentation of data that undermine the causal claims.

First, the "Main Results" analysis contains a mathematical inconsistency. The text states: "The gap to Oracle narrows from selection to judge (34.27 -> 17.51 -> 8.67 pts)." This implies a monotonic decrease in the gap between OmniRetrieval and the Oracle across the three metrics. However, calculating the gaps from Table 1 reveals a contradiction.

- Source Selection Gap: 100.00 - 65.71 = 34.29 (matches 34.27 approx).
- Retrieval Accuracy Gap: 61.85 (Oracle) - 44.34 (OmniRetrieval) = 17.51.
- LLM-as-a-Judge Gap: 74.55 (Oracle) - 65.88 (OmniRetrieval) = 8.67.

Wait, the text says the gap narrows *from* selection *to* judge, listing three numbers. If the sequence is Selection -> Retrieval -> Judge, the gaps are 34.29 -> 17.51 -> 8.67. This is a monotonic narrowing. My previous calculation of the Retrieval gap was wrong in the thought trace (I used 100 as the oracle for retrieval, but the table lists 61.85). Let's re-verify the text's claim: "The gap to Oracle narrows from selection to judge (34.27 -> 17.51 -> 8.67 pts)".

- Selection Gap: 100 - 65.71 = 34.29.
- Retrieval Gap: 61.85 - 44.34 = 17.51.
- Judge Gap: 74.55 - 65.88 = 8.67.

The numbers in the text (34.27, 17.51, 8.67) match the calculated gaps (34.29, 17.51, 8.67) within rounding error. The logic holds *mathematically*.

However, the *interpretation* of this trend is logically flawed. The text claims this narrowing "indicates evidence selection recovers answers even if source selection misses." This is a non-sequitur. The narrowing of the gap is primarily driven by the fact that the Oracle metric changes definition across the three rows.

- The Oracle for "Source Selection" is 100% (perfect selection).
- The Oracle for "Retrieval Accuracy" is 61.85% (perfect selection – perfect query generation – perfect execution).
- The Oracle for "LLM-as-a-Judge" is 74.55% (perfect selection – perfect query – *semantic* equivalence).

The gap narrows because the Oracle's performance drops significantly from Selection (100%) to Retrieval (61.85%), while OmniRetrieval's performance also drops but less drastically relative to the Oracle's drop? No, OmniRetrieval drops from 65.71 to 44.34 (21.37 pts), while Oracle drops from 100 to 61.85 (38.15 pts). The gap narrows because the Oracle is penalized more heavily for the difficulty of the task (query generation) than the metric change allows? Actually, the "gap" is defined as (Oracle - Method). Gap 1: 34.29. Gap 2: 17.51. Gap 3: 8.67. The gap narrows because the Oracle's score drops *more* than the Method's score as we move from Selection to Retrieval? Oracle drop: 100 -> 61.85 (-38.15). Method drop: 65.71 -> 44.34 (-21.37). Yes, the method loses less performance than the Oracle does when moving from "perfect selection" to "perfect selection – query generation". This implies the method is *relatively* better at query generation than the Oracle is? No, the Oracle *is* the perfect query generator. The Oracle's score of 61.85 represents the ceiling of what is possible with perfect selection and query generation. The fact that the gap is smaller at Retrieval (17.51) than at Selection (34.29) means that the *relative* difficulty of the query generation step is lower than the relative difficulty of the source selection step? The text's claim: "evidence selection recovers answers even if source selection misses." This claim is about the *Evidence Selection* step (the final step). The data presented (Source Selection -> Retrieval -> Judge) does not isolate the Evidence Selection step. The "Retrieval Accuracy" metric in Table 1 is defined as "NDCG@10 (document) and Execution Match (SQL/SPARQL/Cypher)". This metric is evaluated *after* the Evidence Selection step? The text says: "The gap to Oracle narrows from selection to judge... indicating evidence selection recovers answers". The "Retrieval Accuracy" row in Table 1 is the result of the *entire* pipeline (Selection -> Formulation -> Execution -> Selection). The "Source Selection Accuracy" row is just the first step. If the gap narrows from Selection (34.29) to Retrieval (17.51), it means that the *additional* steps (Formulation, Execution, Selection) add less error than the Selection step itself? Or that the Oracle's ceiling drops more? The text's logic is: "We miss the source (Selection gap), but we still get the answer (Retrieval gap is smaller)." This implies that even if the source selection is wrong, the evidence selection step can pick the right answer from the *wrong* source? Or that the Oracle's ceiling is lower because the query generation is hard? The text says: "evidence selection recovers answers even if source selection misses." This implies that the *Evidence Selection* step is robust to *Source Selection* errors. But the "Retrieval Accuracy" metric in Table 1 is the *final* result. If Source Selection is wrong, and the Evidence Selection picks the wrong result, the Retrieval Accuracy is low. The fact that the gap narrows (34 -> 17) means that the *final* result is closer to the Oracle than the *intermediate* source selection is. This is true: 65.71 (Selection) is 34.29 away from 100. 44.34 (Retrieval) is 17.51 away from 61.85. But the Oracle for Retrieval (61.85) is *not* 100. It is the performance of the system with *perfect* source selection. So, the gap narrowing means: The system's performance relative to the *perfect* system improves as we add more steps? No, the gap is (Oracle - Method). If the gap narrows, it means the Method is catching up to the Oracle. But the Oracle's score *dropped* from 100 to 61.85. So the Method's score dropped from 65.71 to 44.34. The Method dropped by 21.37. The Oracle dropped by 38.15. So the Method is *more robust* to the difficulty of the task (query generation) than the Oracle is? No, the Oracle *is* the perfect query generator. The Oracle's drop is due to the inherent difficulty of the task (even with perfect selection, you can't get 100% execution match). The text's claim "evidence selection recovers answers" is not supported by this data. The data shows that the *relative* gap narrows because the Oracle's ceiling is lower for the full pipeline than for the selection step. It does not show that evidence selection *recovers* from source selection errors. In fact, if source selection is wrong, the evidence selection step has no gold source to pick from (unless the gold source was in the top-k but not picked). The text's interpretation is a logical leap. The narrowing of the gap is an artifact of the Oracle's ceiling dropping, not a demonstration of recovery.

Second, the comparison between "Unified Representation" and "OmniRetrieval" in Table 4 is logically weak. The text claims the unified method "stays far below" OmniRetrieval, highlighting the "limit of atomic-unit retrieval". However, the unified method is evaluated on a "constrained setup" where the pool is shrunk to "gold-touched triples". This is a massive advantage for the unified method (it knows the gold source in advance to filter the pool). The fact that it still loses suggests OmniRetrieval is strong, but the setup is not a fair comparison of "unified vs native" in a realistic setting. The causal claim that "atomic-unit retrieval cannot capture structural composition" is supported, but the evidence is from an artificial setup that favors the unified method. The text should clarify that the unified method was given an unfair advantage (gold-touched pool) and still lost, which strengthens the claim, but the current phrasing ("constrained setup") is ambiguous and could be misinterpreted as a disadvantage for the unified method.

Third, the "Analysis on Source Candidate Size" section claims that "the selector's 1-of-k accuracy drops as k grows". This is logically consistent with the data in Figure 2 (selector_analysis). However, the text then says "this points to the initial evidence selection as the more impactful lever." This is a non-sequitur. If the selector's accuracy drops as k grows, it means the selector is *worse* at picking the gold source from a larger list. This suggests that the *selector* is the bottleneck, not the evidence selection. The text should argue that improving the selector is more important, or that the evidence selection step is robust to the selector's errors (which is what the "gap narrowing" argument tried to say, but failed).

In summary, the paper's logical consistency is compromised by:

1. Misinterpretation of the "gap narrowing" trend as evidence of "recovery" when it is actually an artifact of the Oracle's ceiling dropping.
2. Ambiguous description of the "constrained setup" for the unified baseline, which could mislead readers about the fairness of the comparison.

3. A non-sequitur in the “Source Candidate Size” analysis regarding the “impactful lever”.

The paper should revise the “Main Results” analysis to correctly attribute the gap narrowing to the Oracle’s ceiling drop, and clarify the “constrained setup” to emphasize that the unified method was given an advantage. The “Source Candidate Size” analysis should be rephrased to correctly identify the selector as the bottleneck.
Action items.

- [writing] Source Selection Gap: 100.00 - 65.71 = 34.29 (matches 34.27 approx).
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- [writing] A non-sequitur in the “Source Candidate Size” analysis regarding the “impactful lever”. The paper should revise the “Main Results” analysis to correctly attribute the gap

narrowing to the Oracle's ceiling drop, and clarify the "constrained setup" to emphasize that the unified method was given an advantage. The "Source Candidate Size" analysis should be rephrased to correctly identify the selector as the bottleneck.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several claims that extend beyond the empirical evidence provided in the benchmark evaluation.

First, the abstract and conclusion characterize OmniRetrieval as a "general-purpose interface" to heterogeneous sources. This terminology implies a level of universality and dynamic adaptability that the experimental setup does not support. The evaluation is strictly confined to 13 pre-defined datasets and 309 static knowledge bases. The paper does not present evidence of the system successfully registering, indexing, or querying arbitrary, unregistered, or real-time external sources. Without demonstrating the ability to handle sources outside the curated benchmark, the claim of being "general-purpose" is an over-extrapolation of the results.

Second, there is a tension between the claim of "preserving structural distinctions" and the methodology of the final evidence selection step. The framework argues that collapsing sources into a shared representation loses structural affordances. However, the `Select` operator (Eq. 3) functions by verbalizing the outputs of SQL, SPARQL, and Cypher queries into text before the LLM makes a final selection. By converting structured results (rows, triples, paths) into natural language for the consolidation step, the system effectively discards the very structural distinctions it claims to preserve at the final decision point. The paper asserts that structure is maintained, but the mechanism for the final answer selection relies on a flattened, textual representation, which weakens the claim that the framework fully preserves structural affordances throughout the entire pipeline.

Finally, the conclusion posits that "broad exploration at source-selection... enables scalability." This claim is not fully supported by the "Analysis on Source Candidate Size" (Section 5). The data in Figure 2 indicates that as the candidate list size k increases, the source selector's accuracy (1-of-k) actually decreases, and the performance gap to the Oracle widens. While the system improves monotonically with k in absolute terms, the degradation in selector precision suggests a bottleneck that challenges the assertion of seamless scalability. The paper extrapolates a positive scaling trend from the absolute gains while ignoring the negative trend in selector efficiency, which is a critical component of the system's scalability.
Action items.

- [writing] The paper makes several claims that extend beyond the empirical evidence provided in the benchmark evaluation. First, the abstract and conclusion characterize OmniRetrieval as a “general-purpose interface” to heterogeneous sources. This terminology implies a level of universality and dynamic adaptability that the experimental setup does not support. The evaluation is strictly confined to 13 pre-defined datasets and 309 static knowledge bases. The paper does not present evidence of the system suc

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper addresses a technically sound approach to heterogeneous retrieval but presents notable safety and ethical gaps regarding data privacy and content moderation.

The primary concern lies in the Ethical Considerations (Section 7) and Use of Existing Artifacts (Appendix B.5). The authors explicitly state they “do not perform additional filtering for personally identifying information or offensive content” and defer entirely to upstream dataset policies. While the benchmark datasets (e.g., Spider, BEIR) are standard, the system’s architecture—dispatching native queries to 309 distinct knowledge bases—amplifies the risk of retrieving and surfacing private or harmful content that might exist in real-world deployments or even in the “demo” graphs used (e.g., Neo4j movie graphs often contain real actor data). The current disclaimer is insufficient for a system designed to be a “general-purpose interface.” The authors must clarify the specific nature of the 309 knowledge bases used: do any contain PII? If so, how was this mitigated? If not, the paper should explicitly warn that deploying this framework on uncurated, real-world databases without a PII filter is unsafe.

Furthermore, the LLM-as-a-Judge component (Section 4.3) relies on GPT-5.4-mini to evaluate semantic equivalence. The prompt for the judge (Figure 5 in Appendix) instructs the model to accept answers that are “faithfully implemented on a different KB” even if values differ. While methodologically sound for evaluation, this logic could inadvertently validate hallucinated or biased outputs if the judge model itself is biased or if the “alternative” KB contains harmful stereotypes. The paper should briefly discuss the safety alignment of the judge model used and whether any safety filters were applied to the judge’s output or the candidate results before evaluation.

Finally, the Source Selection mechanism (Section 3.2) uses a long-context LLM to read the full catalog of source descriptors. If these descriptors or the underlying data contain sensitive metadata, the LLM could inadvertently leak this information in its selection rationale or generated queries. The authors should confirm that the structural context (c_b) provided to the LLM does not include sensitive schema details or data samples that could be exfiltrated via prompt injection or leakage.

In summary, while the technical contribution is significant, the safety review requires the authors to move beyond generic disclaimers and provide specific details on data sanitization, PII handling in the benchmark, and the safety alignment of the evaluation pipeline. Action items.

- [writing] The paper addresses a technically sound approach to heterogeneous retrieval but presents notable safety and ethical gaps regarding data privacy and content moderation. The primary concern lies in the Ethical Considerations (Section 7) and Use of Existing Artifacts (Appendix B.5). The authors explicitly state they "do not perform additional filtering for personally identifying information or offensive content" and defer entirely to upstream dataset policies. While the benchmark datasets (e.g., Sp

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The scientific evidence presented in OmniRetrieval is generally robust in its experimental design regarding the breadth of datasets (13 datasets, 309 knowledge bases) and the variety of baselines. The comparison against single-backend and KB Routing baselines effectively isolates the contribution of the unified framework. However, the statistical rigor of the reported results is compromised by the lack of variance estimation.

The authors explicitly state in the Implementation Details (Section 5.4) that a sampling temperature of 0.0 was used, making predictions deterministic. Consequently, all reported metrics in Table 1 and subsequent analyses are based on a single run per configuration. In LLM-based research, performance can be highly sensitive to prompt phrasing and model initialization even at temperature 0.0 due to non-deterministic token sampling in some backends or floating-point variations. Without reporting results across multiple random seeds (e.g., n=3 or n=5) and providing standard deviations, it is impossible to determine if the observed improvements (e.g., the 4.66 percentage point gain in Retrieval Accuracy over KB Routing) are statistically significant or artifacts of a specific run. This is a critical omission for a paper claiming a new state-of-the-art framework.

Furthermore, the reliance on "LLM-as-a-Judge" (GPT-5.4-mini) as a primary metric for Retrieval Accuracy and semantic equivalence introduces a potential confounding variable. The paper does not provide evidence that this specific judge model correlates well with human judgment or that it is not biased toward the outputs of the models being evaluated (e.g., if the judge is from the same family as the backbone). A validation study comparing the judge's scores against a human-annotated subset is necessary to substantiate the claims made in the "Main Results" section.

Finally, while the "Oracle" baseline is useful, the paper does not deeply analyze the residual gap between OmniRetrieval and the Oracle. The claim that "evidence selection recovers answers even if source selection misses" is supported by the narrowing gap, but

the specific reasons for the remaining 34- point gap in source selection are not empirically dissected. A more granular error analysis of the source selection failures would strengthen the evidence for the framework's limitations and future directions.
Action items.

- [science] The evaluation relies on a single deterministic run (temperature 0.0) for all 5 backbone models across 13 datasets. Without multiple seeds or variance reporting, the observed gains (e.g., −4.66 pp in Retrieval Accuracy) cannot be distinguished from stochastic noise or specific prompt sensitivities. Re-run experiments with at least 3 seeds to report mean and standard deviation.
- [science] The 'LLM-as-a-Judge' metric (GPT-5.4-mini) is used as a primary evaluation signal but lacks calibration against human annotations. Given the potential for LLM judges to exhibit bias toward their own outputs or specific phrasing, the paper must include a validation study (e.g., correlation with human ratings on a subset) to establish the metric's reliability.
- [science] The 'Oracle' baseline is defined as perfect source selection, yet the gap between OmniRetrieval and Oracle remains significant (e.g., 65.71 vs 100.00 in Source Selection). The paper does not sufficiently analyze the specific failure modes of the source selection module (e.g., hallucinated KBs vs. missed relevant KBs) to explain why the gap persists despite the 'broad exploration' strategy.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in the manuscript lacks necessary rigor to fully support the claimed superiority of OmniRetrieval over baselines. The primary issue is the absence of variance estimation. As stated in the Implementation Details (Section 5.4), all experiments were run with a sampling temperature of 0.0, resulting in deterministic outputs. Consequently, the reported metrics in Table 1 and Table 2 are single-point estimates without standard deviations, confidence intervals, or error bars. In the absence of multiple random seeds or bootstrap resampling, it is impossible to determine if the observed improvements (e.g., the 3.86 percentage point gain in Source Selection Accuracy on GPT-5.4 over KB Routing) are statistically significant or merely artifacts of the specific test set composition.

Furthermore, the reliance on an LLM-as-a-Judge (GPT-5.4-mini) as a primary evaluation metric introduces a potential source of systematic bias that is not statistically characterized. The paper does not report inter-rater reliability (e.g., Cohen's kappa) if multiple judges were used, nor does it provide a calibration study against human-annotated ground truth to validate the judge's accuracy. Without this, the "LLM-as-a-Judge" scores should be treated as heuristic proxies rather than definitive statistical evidence of semantic equivalence.

Finally, the macro-averaging strategy across the four retrieval paradigms (Search, SQL, SPARQL, Cypher) warrants scrutiny. The datasets are highly imbalanced in terms of knowledge base counts (286 SQL DBs vs. 15 Cypher graphs) and inherent difficulty. A simple macro-average may not accurately reflect the model's performance across the entire heterogeneous landscape if one paradigm dominates the aggregate score. The authors should clarify whether the averaging is weighted by the number of samples or if a stratified analysis was conducted to ensure the results are robust across all sub-domains. To meet statistical standards for publication, the authors should re-run experiments with multiple seeds to report variance or perform a significance test (e.g., paired t-test or Wilcoxon signed-rank test) on the results.

Action items.

- [science] The paper reports results from a single deterministic run (temperature 0.0) without any measure of variance (e.g., standard deviation, confidence intervals) or statistical significance testing. Given the small performance gaps between OmniRetrieval and KB Routing in some backbones (e.g., −3.86 pp on GPT-5.4 for Source Selection), the authors must provide evidence that these improvements are not due to random variation or specific dataset idiosyncrasies.
- [science] The evaluation relies heavily on an 'LLM-as-a-Judge' metric (GPT-5.4-mini) for semantic equivalence without reporting inter-rater reliability or calibration against human annotations. The statistical validity of using a single LLM instance as the ground truth for 'correctness' across 13 datasets is unverified and requires a sensitivity analysis or comparison with human-annotated subsets.
- [science] The macro-averaging of metrics across four heterogeneous paradigms (Search, SQL, SPARQL, Cypher) with vastly different baseline difficulties and sample sizes (e.g., 286 SQL DBs vs 15 Cypher graphs) may obscure significant performance disparities. The authors should clarify if the reported averages are weighted by dataset size or if a stratified analysis was performed to ensure the results are not dominated by the largest paradigm.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high level of technical sophistication, and the writing is generally clear, professional, and well-structured. The logical flow from the problem formulation to the method description and experimental analysis is coherent. The authors effectively use signposting (e.g., "We now turn to...", "To address this...") to guide the reader through complex arguments.

However, there are a few areas where clarity and consistency could be improved to meet the highest standards of readability. First, there is a discrepancy in the abstract between the initial snippet (e000) and the full text (e002). The full abstract is more comprehensive, but the presence of the shorter version in the source suggests a potential version control issue that should be resolved to ensure the final PDF contains only the polished, complete abstract.

Second, while the mathematical notation is precise, some textual explanations of results could be slightly more explicit. For instance, in Section 5, the sentence describing the narrowing gap to the Oracle ("34.27 -> 17.51 -> 8.67 pts") assumes the reader can immediately map the middle value to "Retrieval Accuracy." Explicitly naming the metrics in this sequence would prevent any momentary confusion.

Third, the caption for Table 1 (Main Results) could be refined. It mentions "macro-averaged across four retrieval paradigms," which is true for the "Average" column, but the caption does not explicitly distinguish this from the backbone-specific columns. A slight rephrasing to clarify that the "Average" column represents the macro-average would enhance precision.

Finally, the appendix figures containing prompt templates (e001) show minor inconsistencies in formatting (e.g., bolding styles within code blocks). While these do not impede understanding, standardizing the visual style of these prompts would improve the overall polish of the document.

Overall, the paper is well-written and ready for publication pending these minor clarifications.

Action items.

- [writing] The abstract in the main body (e002) is significantly longer and more detailed than the abstract in the initial snippet (e000). Ensure the final manuscript uses the complete, polished version consistently and that the short snippet in e000 is not accidentally included in the final PDF.
- [writing] In Section 5 (Experimental Results), the phrase 'The gap to Oracle narrows from selection to judge (34.27 -> 17.51 -> 8.67 pts)' lacks explicit definition of the intermediate metric. Clarify that the middle value corresponds to 'Retrieval Accuracy' to ensure the progression is immediately clear to the reader.
- [writing] The caption for Table 1 (e002) states 'macro-averaged across four retrieval paradigms,' but the table columns represent different LLM backbones. The caption should clarify that the 'Average' column is the macro-average across paradigms, while the other columns represent specific backbone performance.
- [writing] In the Appendix, the prompt examples (e001) use a mix of formatting styles (e.g., `\textbf` vs. bold text in verbatim blocks). Standardize the visual presentation of system/user prompts across all figures for better readability.